

AI Exposure and Age-Differentiated Employment

Evidence from Norwegian Register Data

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Motivation

AI and the future of work: in the headlines



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The New York Times

A.I. Doesn't Have to Mean Layoffs

Illustration by The Atlantic. Sources: Bryan Hill / Getty; James Mitchell / Getty; Valerii Lelikhov / Getty.

JUNE 2, 2024

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Recent literature: AI exposure and employment

Country	Paper	Headline (youngest workers, top quintile)
United States	Brynjolfsson, Chandar & Chen (2025)	Q5 ages 22–25 decline ~ 16 log pts
Sweden	Lodefalk et al. (2026)	Q5 ages 22–25 decline $\sim 5.5\%$; monotonic gradient
United Kingdom	Teeselink (2025)	-0.3% per SD LLM (large language model) exposure; concentrated in junior, high-wage
39 countries	Teeselink (2026)	Hiring declines; sharper under strict employment protection legislation
Denmark	Humlum & Vestergaard (2026)	Null at firm level: workplace adoption rules out effects $> 1/3$ ppt on early-career share
Finland	Kauhanen et al. (2024, 2025, 2026)	Null; demographic explanation

Contributions

- Complement the international literature on AI exposure and employment pre/post ChatGPT with results on Norway
- Study the recent impact of **Agentic AI** (*“autonomous LLM-based systems that plan, use tools, and execute multi-step research tasks”* (Korinek, 2025))
- Track developments of the universe of Norwegian employment “live”

Data

Data: A-ordningen

- Norway's mandatory employer-reporting system; universe of formal employment.
- Coverage: ~ 3.1 million workers per month, January 2021 through February 2026.
- Variables used: 4-digit STYRK-08 occupation, age (monthly precision), institutional sector, cash earnings (*kontantlønn*), position percentage, contracted hours, new-job flag.
- Two analysis units:
 - **Cell counts and means** (occupation $\times$ age $\times$ month $\times$ sector) for the descriptive series.
 - **Individual records** (firm $\times$ age $\times$ occupation $\times$ month) for the firm-FE design.
- Index date: October 2022 = 1, the month before ChatGPT's public release.

Constructing exposure quintiles

- Rank the 397 STYRK-08 codes that the Eloundou et al. mapping covers by β .
- Assign each 4-digit code to one of five quintiles on the *occupation* distribution.
- Each code counts once, regardless of its employment share.
- Quintile assignment is fixed throughout the period.

Focus on exposure measure of Eloundou et al.

- **Commonly used, standard measure measure.** Published in *Science* (2024), used by Brynjolfsson, Chandar & Chen (2025), later by Teeselink (2025), Humlum & Vestergaard (2026), Kauhanen and Petri (2021), Lodefalk et al. (2026).
- **Best validation against observed AI use.** In Lindenlaub, Oh, Rodríguez & Veldkamp (2026, Fig. 1), Eloundou et al. has $R^2 = 0.30$ against worker-reported adoption, ahead of Handa (0.26), Felten (0.26), and Webb (0.02).

Show Lindenlaub et al. Fig. 1

Mapping from US to Norwegian occupation codes

- Eloundou et al.'s exposure measure is built on US occupation codes (SOC).
- Norwegian register data use STYRK-08.
- STYRK-08 $\equiv$ ISCO-08 at 4 digits (Statistics Norway / SSB, Notater 17/2011).
- Bridge: SOC $\rightarrow$ ISCO-08 via US Bureau of Labor Statistics (BLS) crosswalk (Aug 2012, rev. June 2015).

O*NET: the US occupation database underneath three of the four measures

- Maintained by the US Department of Labor; covers $\sim 1,000$ occupations on SOC 2018 codes (O*NET-SOC 2019 taxonomy)
- Each occupation has structured content:
 - **Tasks:** discrete activities ($\sim 19,000$ across all occupations; Pharmacists: 21 tasks, e.g. “review prescriptions for accuracy”). Tagged *Core* or *Supplemental*; no numerical weight.
 - **Abilities:** 52 underlying attributes (e.g. Deductive Reasoning, Manual Dexterity). Scored on prevalence and importance per occupation.
 - Also: skills, knowledge, work activities, work context (not used by the four measures here).
- Eloundou et al. and Handa operate on **tasks**; Felten operates on **abilities**; Anthropic 2026 operates at the occupation level.

What Eloundou et al. measure

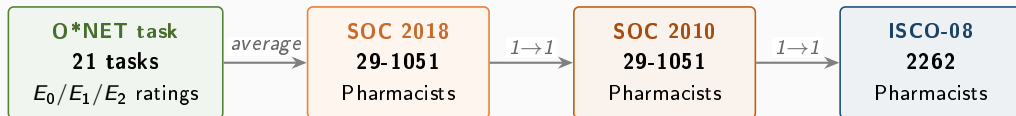
- Each O*NET task rated by humans and GPT-4 on three categories:
 - E_0 no exposure
 - E_1 LLM alone reduces task time by $\geq 50\%$
 - E_2 LLM with complementary software reduces task time by $\geq 50\%$
- Task-level score: $\beta_{\text{task}} = 0$ if E_0 , 1 if E_1 , 0.5 if E_2
- Occupation score: $\beta_{\text{occ}} = \text{unweighted mean of } \beta_{\text{task}}$ across the occupation's tasks
- O*NET tags tasks as Core / Supplemental; Eloundou et al. do not re-weight by either

The code chain



ISCO-08 = STYRK-08 at 4 digits (SSB Notater 17/2011). STYRK 2262 is
Farmasøyster.

Worked example: Pharmacists (one-to-one mapping)



- Pharmacists carry the same 6-digit SOC in both vintages: 29-1051
- BLS crosswalk sends 29-1051 to a single ISCO-08 code (2262)
- Norwegian STYRK 2262 is Farmasøyster, no Norway-specific re-coding

Task-level scoring for Pharmacists (SOC 29-1051, all 21 tasks)

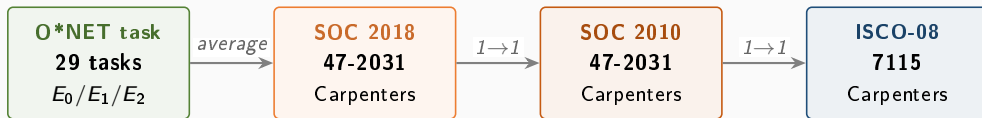
O*NET task	E	β	O*NET task	E	β
Review prescriptions for accuracy	E_2	0.5	Refer patients to other professionals	E_2	0.5
Assess identity, strength, purity of meds	E_0	0.0	Work in hospitals/clinics or for HMOs	E_2	0.5
Provide info on drug interactions	E_2	0.5	Update or troubleshoot pharmacy databases	E_1	1.0
Analyze prescribing trends	E_2	0.5	Manage pharmacy operations, hire, supervise	E_2	0.5
Maintain pharmacy files and patient profiles	E_2	0.5	Prepare sterile solutions	E_0	0.0
Collaborate with other health professionals	E_2	0.5	Offer health promotion / prevention activities	E_0	0.0
Plan procedures for mixing and labeling	E_0	0.0	Assay radiopharmaceuticals	E_0	0.0
Order and purchase pharmaceutical supplies	E_2	0.5	Publish educational information	E_1	1.0
Compound and dispense medications	E_0	0.0			
Contact insurance companies on billing	E_2	0.5	Counts: 6 E_0 3 E_1 12 E_2		
Advise customers on medication brands	E_2	0.5			
Teach pharmacy students serving as interns	E_1	1.0	Occupation β (mean of 21): 0.4286 (Q4)		
Provide services for diabetes, asthma, etc.	E_2	0.5			

Other one-to-one mappings (large Norwegian occupations)

STYRK	Norwegian title	SOC 2010 (US title)	employed	β_{occ}	Q
7115	Tømrere og snekkere	47-2031 Carpenters	47,106	0.144	2
2330	Lektorer (videregående skole)	25-2031 Secondary teachers	31,155	0.333	3
2611	Jurister og advokater	23-1011 Lawyers	11,808	0.425	4
2631	Forskere/rådgivere, samf.økonomi	19-3011 Economists	1,282	0.553	5
3511	Driftsteknikere, IKT	43-9011 Computer Operators	16,517	0.736	5

All five are clean one-to-one in the BLS crosswalk. Exposure spans Q2 to Q5: carpenters at the bottom, ICT operations technicians at the top. SOC titles do not always match the Norwegian title literally (e.g. “Computer Operators” vs. “Driftsteknikere, IKT”).

Worked example: Tømrere og snekkere (STYRK 7115)

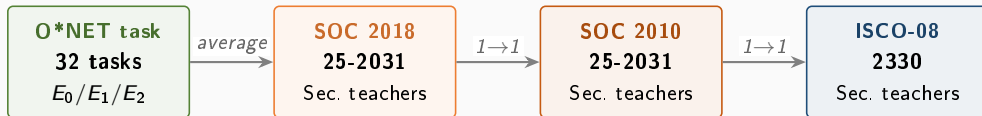


STYRK 7115 is Tømrere og snekkere. Same 6-digit SOC in both vintages, one O*NET specialty, 29 tasks. Mostly E_0 : physical work dominates.

Task-level scoring: Carpenters (SOC 47-2031, 29 tasks)

O*NET task	E	β	O*NET task	E	β
Follow safety rules and regulations	E_0	0.0	Dig or direct digging of post holes	E_0	0.0
Measure and mark cutting lines on materials	E_0	0.0	Cover subfloors with building paper	E_0	0.0
Assemble and fasten materials	E_0	0.0	Construct forms or chutes for concrete	E_0	0.0
Study specifications in blueprints	E_2	0.5	Arrange for subcontractors	E_2	0.5
Shape or cut materials to measurements	E_0	0.0	Build or repair cabinets, doors, floors	E_0	0.0
Verify trueness with plumb bob and level	E_0	0.0	Finish surfaces of woodwork or wallboard	E_0	0.0
Inspect ceiling/floor tile, wall coverings	E_2	0.5	Select and order lumber and materials	E_2	0.5
Erect scaffolding or ladders	E_0	0.0	Work with or remove hazardous material	E_0	0.0
Install windows, frames, fixtures	E_0	0.0	Fill cracks or defects in plaster	E_0	0.0
Maintain records, present written reports	E_1	1.0	Prepare cost estimates for clients	E_2	0.5
Remove damaged or defective parts	E_0	0.0	Perform minor plumbing or concrete work	E_0	0.0
Maintain job records, schedule work crew	E_2	0.5	Apply shock-absorbing or sound-deadening	E_0	0.0
Anchor and brace forms and structures	E_0	0.0	Examine structural timbers for decay	E_0	0.0
Bore boltholes in timber/masonry/concrete	E_0	0.0	Build sleds from logs and timbers	E_0	0.0
Install rough door/window frames, subflooring	E_0	0.0			
Counts: 22 E_0 1 E_1 6 E_2 SOC β (mean of 29 tasks): 0.138 (Q2)					

Worked example: Lektorer videregående (STYRK 2330)

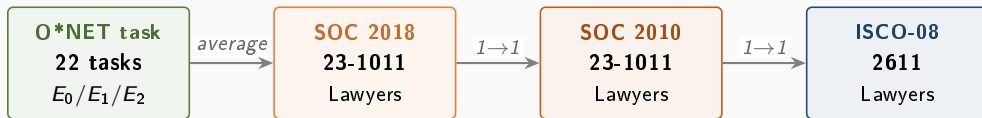


STYRK 2330 is Lektorer mv. (videregående skole). One O*NET specialty, 32 tasks. Most are E_2 (LLM with software is enough) and reflect classroom and administrative work.

Task-level scoring: Secondary teachers (SOC 25-2031, 32 tasks)

O*NET task	E	β	O*NET task	E	β
Instruct through lectures, discussions	E_2	0.5	Plan and conduct field trips	E_0	0.0
Establish rules of student behavior	E_0	0.0	Sponsor extracurricular activities	E_0	0.0
Prepare materials and classrooms	E_2	0.5	Perform administrative duties	E_2	0.5
Adapt teaching to varying abilities	E_2	0.5	Prepare and implement remedial programs	E_2	0.5
Observe and evaluate students' performance	E_2	0.5	Attend professional meetings and workshops	E_0	0.0
Establish clear objectives for lessons	E_2	0.5	Use computers and audiovisual aids	E_2	0.5
Prepare students for later grades	E_2	0.5	Select, store, order, or use class materials	E_2	0.5
Prepare, administer, and grade tests	E_2	0.5	Collaborate with other teachers	E_2	0.5
Maintain accurate, complete records	E_2	0.5	Plan and supervise projects, field studies	E_2	0.5
Enforce attendance and progression rules	E_0	0.0	Meet with professionals on programs	E_0	0.0
Instruct and monitor use of equipment	E_0	0.0	Guide and counsel students	E_2	0.5
Prepare content for diverse-needs students	E_2	0.5	Confer with parents on progress	E_0	0.0
Assign and grade class work and homework	E_2	0.5	Meet with administrators on instruction	E_2	0.5
Prepare reports on student progress	E_2	0.5	Prepare for assigned classes	E_2	0.5
Keep informed about developments in education	E_1	1.0	Provide disabled students with devices	E_2	0.5
Prepare objectives and outlines	E_2	0.5	Administer standardized tests	E_0	0.0
Counts: 12 E_0 1 E_1 19 E_2 SOC β (mean of 32 tasks): 0.328 (Q3)					

Worked example: Jurister og advokater (STYRK 2611)

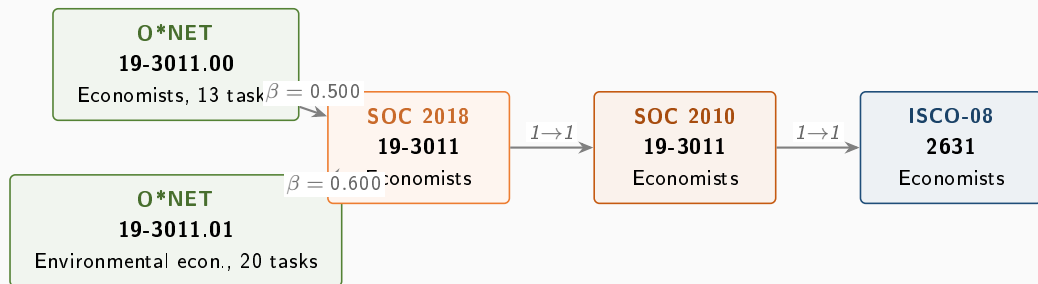


STYRK 2611 is Jurister og advokater. One O*NET specialty, 22 tasks. Nearly all tasks are E_2 : text-heavy work where LLM plus software can substitute.

Task-level scoring: Lawyers (SOC 23-1011, 22 tasks)

O*NET task	E	β	O*NET task	E	β
Advise clients on legal rights	E_2	0.5	Confer with colleagues with specialties	E_0	0.0
Represent clients in court or in hearings	E_0	0.0	Examine legal data for advisability	E_2	0.5
Select jurors, argue motions, meet judges	E_0	0.0	Search and examine public records	E_2	0.5
Study Constitution, statutes, decisions	E_2	0.5	Act as trustee, guardian, or executor	E_2	0.5
Prepare legal briefs and opinions	E_2	0.5	Help establish company legal policies	E_2	0.5
Interpret laws, rulings, regulations	E_2	0.5	Negotiate settlements of civil disputes	E_2	0.5
Prepare and draft legal documents	E_2	0.5	Probate wills, represent estate executors	E_2	0.5
Present and summarize cases to judges	E_2	0.5	Work in environmental, public health law	E_2	0.5
Gather evidence by interviewing clients	E_2	0.5	Perform administrative and management	E_2	0.5
Analyze the probable outcomes of cases	E_2	0.5	Supervise legal assistants	E_2	0.5
Evaluate findings; develop defense strategy	E_2	0.5	Provide pro bono services	E_2	0.5
Counts: 3 E_0 0 E_1 19 E_2 SOC β (mean of 22 tasks): 0.432 (Q4)					

Worked example: Forskere/rådgivere, samf.økonomi (STYRK 2631)

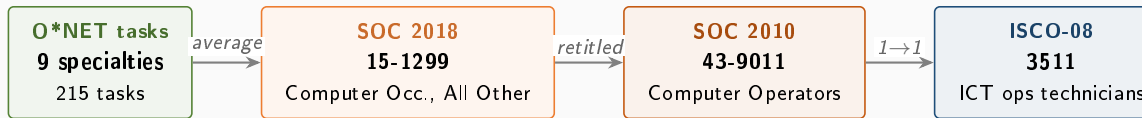


Two O*NET specialties roll up to one SOC 2018 code. The SOC-level β averages the two specialty means: $(0.500 + 0.600)/2 = 0.550$, regardless of how many tasks each specialty contains.

Task-level scoring: Economists (SOC 19-3011, 33 tasks across 2 specialties)

19-3011.00 Economists (13 tasks)			19-3011.01 Env. Economists (20 tasks)		
Study economic and statistical data	E_2	0.5	Prepare and deliver presentations	E_2	0.5
Conduct research on economic issues	E_2	0.5	Develop programs or policy recommendations	E_2	0.5
Compile, analyze, report data	E_2	0.5	Develop economic models and forecasts	E_2	0.5
Supervise research projects	E_2	0.5	Demonstrate economic benefits of policies	E_2	0.5
Teach theories and methods of economics	E_2	0.5	Conduct research on environmental relations	E_2	0.5
Study impacts of public policies	E_2	0.5	Perform complex mathematical analyses	E_1	1.0
Formulate recommendations and plans	E_2	0.5	Write social, legal, economic impact reports	E_1	1.0
Explain economic impact of policies	E_2	0.5	Teach courses in environmental economics	E_2	0.5
Provide advice and consultation	E_2	0.5	Develop programs to promote sustainability	E_2	0.5
Forecast production and consumption	E_2	0.5	Develop systems for analyzing data	E_2	0.5
Develop economic guidelines and standards	E_2	0.5	Write research proposals and grants	E_1	1.0
Testify at regulatory or legislative hearings	E_2	0.5	Examine exhaustibility of natural resources	E_2	0.5
Provide litigation support	E_2	0.5	Monitor or analyze market and env. trends	E_2	0.5
Write technical documents or articles	E_1	1.0	Develop environmental research project plans	E_2	0.5
			Conduct research on economic, env. topics	E_2	0.5
			Collect and analyze data on env. impact	E_2	0.5
			Assess costs and benefits of activities	E_2	0.5
			Identify environmentally friendly business	E_2	0.5
			Interpret indicators of ecosystem health	E_2	0.5
			Collaborate with researchers across fields	E_2	0.5
Specialty mean: 0.500			Specialty mean: 0.600		
SOC $\beta = (0.500 + 0.600)/2 = 0.550$ (Q5)					

Worked example: Driftsteknikere, IKT (STYRK 3511)

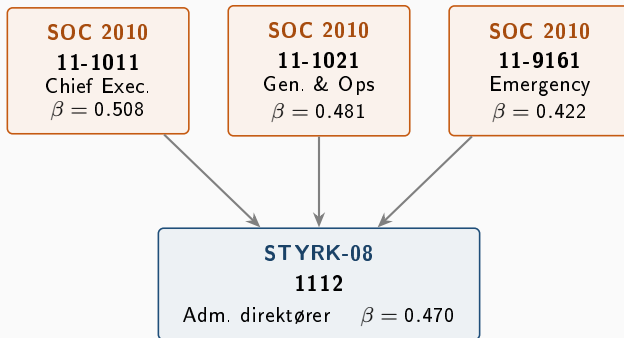


The SOC 2010 → 2018 re-coding turned “Computer Operators” into the catch-all “Computer Occupations, All Other” (15-1299), which contains nine sub-specialties. The Eloundou et al. pipeline averages those nine specialty means; tasks are not aggregated directly.

Specialty-level scoring: Computer Occupations, All Other (15-1299)

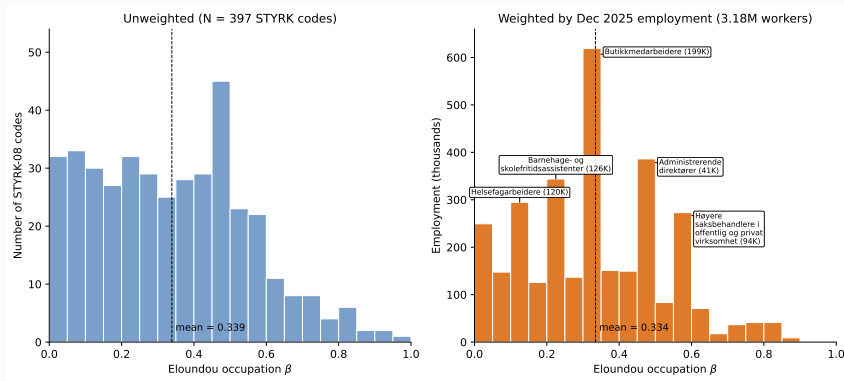
O*NET-SOC	Title	N tasks	specialty β
15-1299.01	Web Administrators	23	0.913
15-1299.02	Geographic Information Systems Technologists	27	0.593
15-1299.03	Document Management Specialists	18	0.611
15-1299.04	Penetration Testers	22	0.795
15-1299.05	Information Security Engineers	20	0.675
15-1299.06	Digital Forensics Analysts	20	0.725
15-1299.07	Blockchain Engineers	17	0.971
15-1299.08	Computer Systems Engineers/Architects	28	0.768
15-1299.09	Information Technology Project Managers	21	0.571
SOC β (mean of 9 specialties): 0.736 (Q5)			

Worked example: many-to-one mapping



- Three SOC 2010 codes map to ISCO 1112 in the BLS crosswalk
- STYRK β is the unweighted mean of contributing SOC-level β
- 57.7% of Eloundou et al. STYRK codes have at least one contributor flagged as a partial match in the BLS file

Distribution of Eloundou et al. β across Norwegian occupations



Left: one observation per STYRK-08 code (397 codes). Right: weighted by December 2025 employment (3.19M workers); five tallest spikes labelled. Means nearly identical; weighted mass concentrates in a handful of large mid-skill occupations.

Five largest occupations per exposure quintile (Apr 2025)

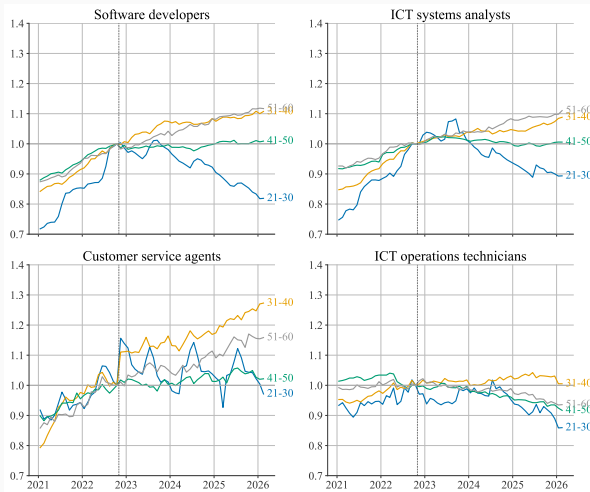
Occupation	Score	Emp.
<i>Q1 (least exposed)</i>		
Health care assistants	0.11	115k
Cleaners (offices, hotels, ...)	0.03	70k
Building electricians	0.13	31k
Food machine operators	0.09	26k
Building caretakers	0.00	24k
<i>Q2</i>		
Child care workers	0.25	111k
Personal care, health (n.e.c.)	0.22	78k
Carpenters and joiners	0.14	43k
Early childhood educators	0.23	35k
Home-based personal care	0.22	23k
<i>Q3</i>		
Shop sales assistants	0.34	131k
Primary school teachers	0.33	89k
(STYRK 2223)	0.34	67k
Stock clerks	0.33	35k

Occupation	Score	Emp.
<i>Q4</i>		
Managing directors / CEOs	0.49	40k
University, higher-ed teachers	0.49	39k
Retail/wholesale trade managers	0.48	35k
Civil engineers	0.48	28k
Business admin managers (n.e.c.)	0.49	28k
<i>Q5 (most exposed)</i>		
Policy administration profs.	0.57	91k
General office clerks	0.60	60k
Commercial sales representatives	0.57	45k
Systems analysts	0.63	33k
Software developers (n.e.c.)	0.77	22k

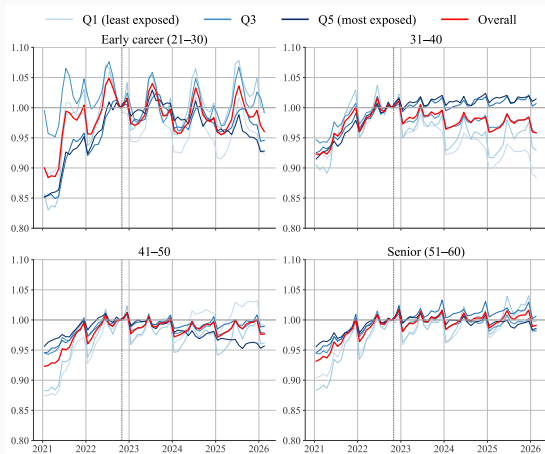
Employment in thousands. ISCO-08 labels. Ranked within quintile by April 2025 headcount across both sectors, ages 21–60.

Descriptive results

Selected high-exposure occupations



Private sector: employment by exposure quintile and age



No clean negative gradient at the youngest ages. A gap opens at ages 41–50 after 2022.

Cell-level difference-in-differences

Empirical specification

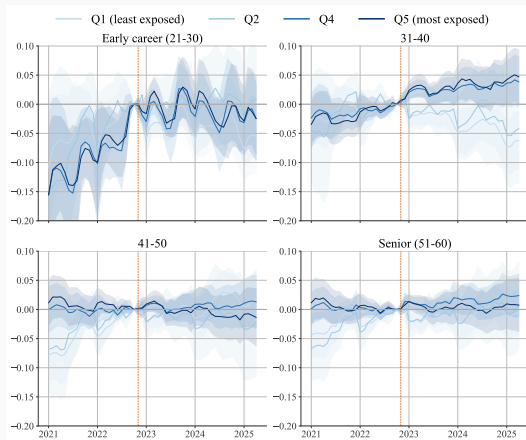
- Cells: (occupation $j \times$ age group $a \times$ month t), private sector
- One Poisson PPML regression per decade age group:

$$\log E[\text{count}_{j,a,t}] = \alpha_j + \beta_t + \gamma_{q(j),k(t)}$$

- α_j : occupation FE, absorbs permanent occupation size
- β_t : month FE, absorbs the aggregate age-cohort time path
- $\gamma_{q,k}$: log-point change for quintile q at event time k , relative to **Q3** (median exposure) and $k = -1$ (October 2022)
- SE clustered at occupation

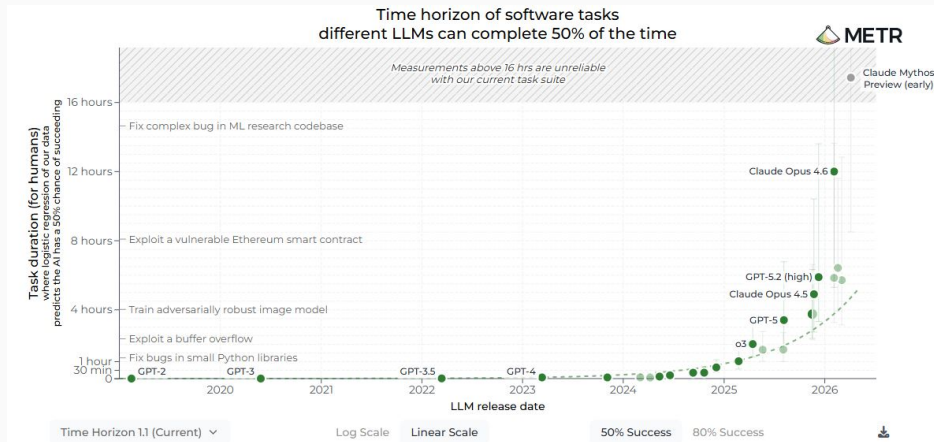
Why Q3 reference? Q1 (least exposed) is dominated by winter-construction seasonality. Q3 is the median exposure, less noise baseline. **Why Poisson?** Counts with some zeros; $\log(0 + x)$ biased.

Cell-level event-study grid (Q3 reference)



Q1, Q2, Q4, Q5 plotted against Q3. Cell-level Poisson event-study coefficients by decade age group, private sector. Reference: October 2022 ($k = -1$).

AI capability is accelerating



Length of software tasks an LLM completes at 50 % success has roughly doubled every ~ 7 months since GPT-2. By 2026, the frontier handles multi-hour tasks. Source: Model Evaluation & Threat Research (METR).

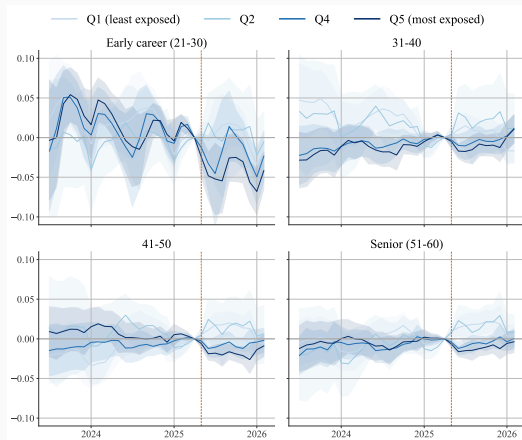
Why agentic AI is a separate window

Mechanism. Pre-agentic AI: AI helps with particular steps of a production chain, humans verify each step. Agentic AI: humans verify only the final output of a chain (Demirer, Horton, Immorlica & Lucier 2026).

When did this happen? Somewhat arbitrary, user cost falls gradually.

- **Early 2024.** Devin demo (Mar 2024), Claude 3 Opus (Mar 2024), standardised function-calling APIs. Chaining technically possible but demo-grade.
- **May 2025.** OpenAI Codex (16 May 2025) and Anthropic Claude 4 / Claude Code (22 May 2025) make multi-step coding agents production-grade.

Re-anchor to agentic AI (April 2025)



Same Poisson DiD, re-anchored to April 2025 (last pre-agentic month). Pre-period 2023m1–2025m4, post 2025m5–2026m2. Cell-level, private sector. Q3 reference.

Conclusion

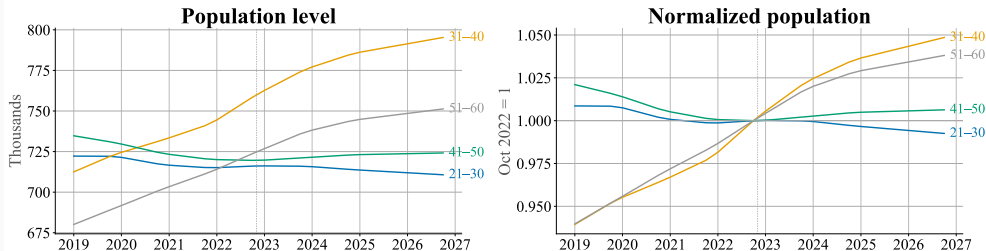
Takeaway

- Pre/post ChatGPT (Oct 2022). ...
- Re-anchor to agentic AI (Apr 2025).
- All results in the **private sector**; ...

Robustness

Population and cohort adjustment

Population by age group: Norway 2021–2026



Source: SSB table 07459 (population by single year of age, January 1). Quarterly values interpolated linearly.

- Per-capita normalisation (divide by SSB resident population in the same age cell).
- Composition adjustment fixes the within-bin age distribution at 2021-Q1.
- The age-by-quintile pattern is unchanged after both adjustments.

Backup

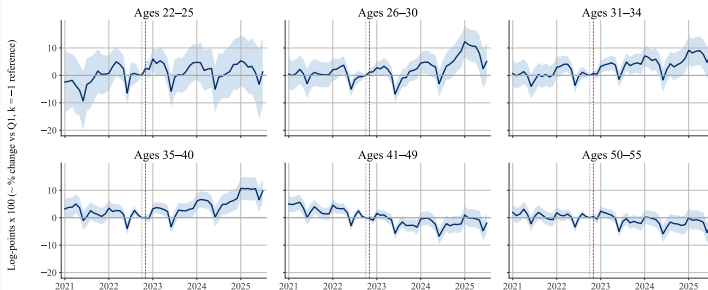
Validation: within-firm reallocation

- The cell-level gradient could reflect compositional shifts across firms rather than within-firm reallocation.
- We re-estimate on individual-level register data with firm $\times$ time fixed effects, as in Brynjolfsson–Chandar–Chen.
- Individual-level data is a snapshot; we use it to validate the cell-level pattern, not for time-tracking.

Specification. Poisson PPML with firm $\times$ age, firm $\times$ month, age $\times$ month fixed effects. Triple-difference on $\text{Young}_a \times \text{Post}_t \times \text{Exposure}_q$. Sample: private *foretak*, ≥ 20 workers in ages 22–55, 2021m1–2025m7. SE clustered at *foretak*.

Event study by age bin (legacy: Q5 vs Q1)

Employment of Q5 (most AI-exposed) occupations vs. Q1, by age cohort
firm-FE event study, Poisson



Sample: private-sector foretak (sekt = 3), 22–55-year-olds, 2021m1–2025m7. Specification: $\text{fepois}(\text{count_all} - i(k, \text{ai_q}, \text{ref} = -1, \text{ref2} = 1) \mid \text{foretak}^q + \text{foretak}^{\text{month}})$, clustered at foretak, estimated separately per age bin. Reference: $k = -1$ (October 2022) and $q = 1$ (least exposed). Dashed red line: November 2022 (GPT-4 era). Tick marks at January of each year. Shaded band: 95 % CI. Poisson log-point coefficients $\gamma_{\{Q5,k\}}$ multiplied by 100 read approximately as percent change in Q5 employment relative to Q1 employment, at event time k vs the omitted reference month. Source: from `_secure_server/coefficients/coef_event_study_fepois.csv`.

22–25 flat with wide CIs. Ages 26–30, 31–34, 35–40 drift up to +8 to +10 log points by mid-2025. Ages 41–49 and 50–55 are flat to slightly negative (–3 to –5). Reference $k = -1$ (October 2022).

Lindenlaub et al. (2026), Fig. 1: AI adoption vs. four exposure measures

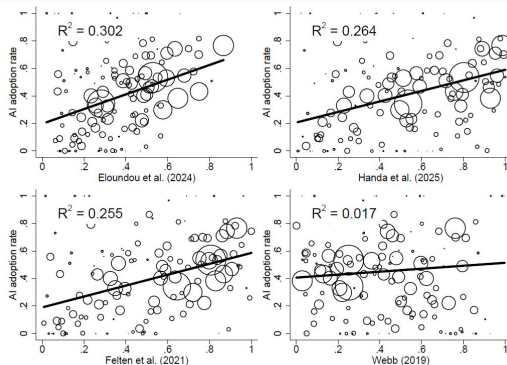
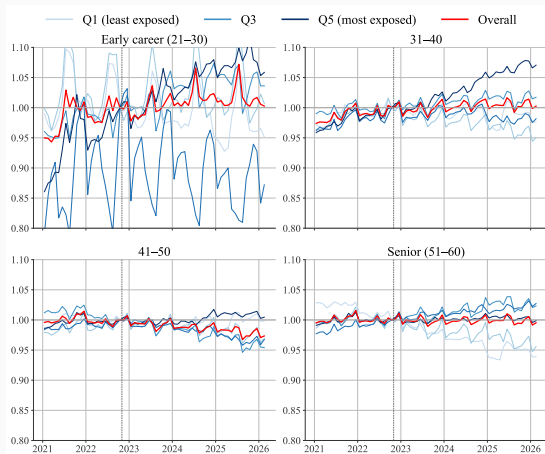


Figure 1: Relation between Actual Adoption by Workers and AI Exposure Measures

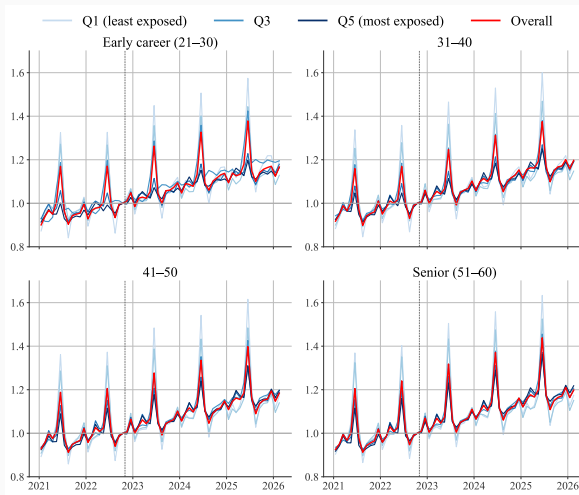
Each panel regresses worker-reported AI adoption on an exposure measure. Eloundou et al. has $R^2 = 0.30$, ahead of Handa (0.26), Felten (0.26), and Webb (0.02).

Public sector: no exposure gradient



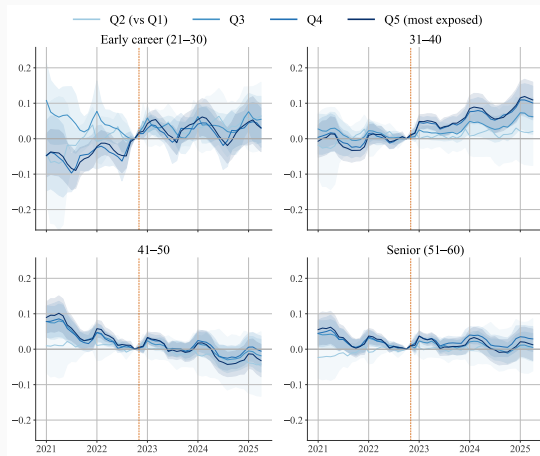
Same indexed series as the private-sector slide, public sector. No exposure gradient at any age.

Cash earnings: no divergence



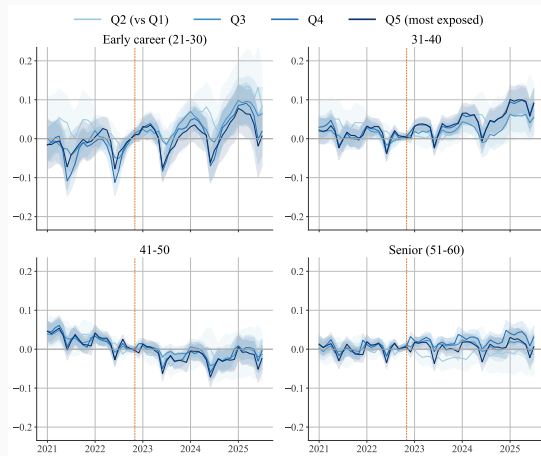
Cash earnings (*kontantlønn*) by quintile and age in the private sector. Consistent with adjustment on the hiring margin under two-tier wage bargaining.

Cell-level event study, Q1 reference



Same cell-level Poisson event study, plotted against Q1 (least exposed) instead of Q3. Magnitudes are larger but the qualitative pattern is the same.

Firm-FE event study, full quintile grid



Q2–Q5 vs Q1 event-study coefficients by age bin (Poisson, firm-FE).